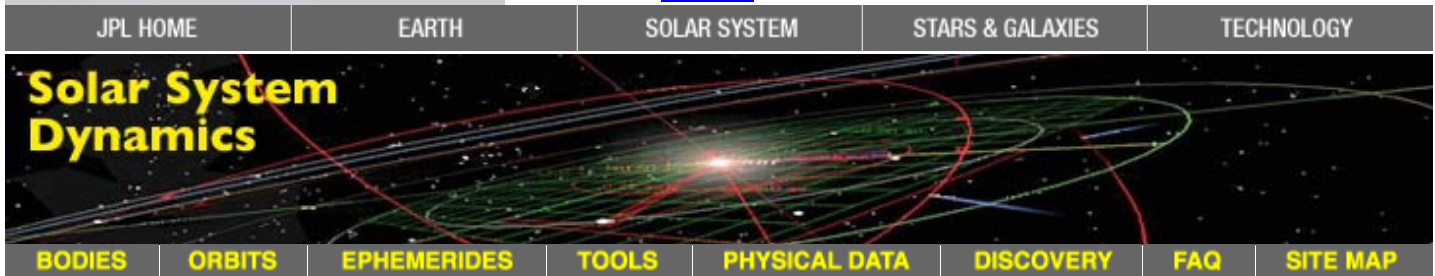




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Planetary Satellite Physical Parameters

The following tables provide known physical parameters for [planetary satellites](#). [References](#), where known, are shown in square brackets (e.g. [1]) following the particular value. The [JPL lunar constants and models technical document](#) is also available.

Earth's Moon

Sat.	<u>GM</u> (km ³ /sec ²)	<u>Mean radius</u> (km)	<u>Mean density</u> (g/cm ³)	<u>Magnitude</u> V_0 or R	<u>Geometric Albedo</u>
Moon	4902.801±0.001 [1]	1737.5±0.1 [2]	3.344±0.005	-12.74 [3]	0.12 [3]

Martian System

Sat.	<u>GM</u> (km ³ /sec ²)	<u>Mean radius</u> (km)	<u>Mean density</u> (g/cm ³)	<u>Magnitude</u> V_0 or R	<u>Geometric Albedo</u>
Phobos	0.0007112±0.0000010 [4]	11.1±0.15 [5]	1.872±0.076	11.4±0.2 [6]	0.071±0.012 [6]
Deimos	0.0000985±0.0000024 [4]	6.2±0.18 [5]	1.471±0.166	12.45±0.05 [6]	0.068±0.007 [7]

Jovian System

Sat.	<u>GM</u> (km ³ /sec ²)	<u>Mean radius</u> (km)	<u>Mean density</u> (g/cm ³)	<u>Magnitude</u> V_0 or R	<u>Geometric Albedo</u>
Io	5959.916±0.012 [8]	1821.6±0.5 [9]	3.528±0.006	5.02±0.03 [14]	0.63±0.02 [16]
Europa	3202.739±0.009 [8]	1560.8±0.5 [10]	3.013±0.005	5.29±0.02 [14]	0.67±0.03 [17]
Ganymede	9887.834±0.017 [8]	2631.2±1.7 [11]	1.942±0.005	4.61±0.03 [14]	0.43±0.02 [18]
Callisto	7179.289±0.013 [8]	2410.3±1.5 [12]	1.834±0.004	5.65±0.10 [14]	0.17±0.02 [18]

Amalthea	0.138±0.030	[8]	83.45±2.4	[13]	0.849±0.199	14.1±0.2	[15]	0.090±0.005	[19]
Himalia	0.45		85	[20]	2.6	14.2R	[26]	0.04	
Elara	0.058		43	[20]	2.6	16.0R	[26]	0.04	
Pasiphae	0.020		30	[20]	2.6	16.8R	[26]	0.04	
Sinope	0.0050		19	[20]	2.6	18.2R	[26]	0.04	
Lysithea	0.0042		18	[20]	2.6	18.1R	[26]	0.04	
Carme	0.0088		23	[20]	2.6	18.1R	[26]	0.04	
Ananke	0.0020		14	[20]	2.6	19.1R	[26]	0.04	
Leda	0.00073		10	[20]	2.6	19.2R	[26]	0.04	
Thebe	0.10		49.3±2.0	[21]	3.0	16.0	[22]	0.047±0.003	[19]
Adrastea	0.0005		8.2±2.0	[21]	3.0	18.7	[22]	0.1±0.045	[21]
Metis	0.008		21.5±2.0	[21]	3.0	17.5	[22]	0.061±0.003	[19]
Callirrhoe	0.000058		4.3	[25]	2.6	20.8R	[26]	0.04	[25]
Themisto	0.000046		4.0	[25]	2.6	21.0R	[26]	0.04	[25]
Megaclete	0.000014		2.7	[25]	2.6	21.7R	[26]	0.04	[25]
Taygete	0.000011		2.5	[25]	2.6	21.9R	[26]	0.04	[25]
Chaldene	0.000005		1.9	[25]	2.6	22.5R	[25]	0.04	[25]
Harpalyke	0.000008		2.2	[25]	2.6	22.2R	[26]	0.04	[25]
Kalyke	0.000013		2.6	[25]	2.6	21.8R	[26]	0.04	[25]
Iocaste	0.000013		2.6	[25]	2.6	21.8R	[26]	0.04	[25]
Erinome	0.000003		1.6	[25]	2.6	22.8R	[25]	0.04	[25]
Isonoe	0.000005		1.9	[25]	2.6	22.5R	[25]	0.04	[25]
Praxidike	0.000029		3.4	[25]	2.6	21.2R	[26]	0.04	[25]
Autonoe	0.000006		2.0	[25]	2.6	22.0R	[25]	0.04	[25]
Thyone	0.000006		2.0	[25]	2.6	22.3R	[25]	0.04	[25]
Hermippe	0.000006		2.0	[25]	2.6	22.1R	[25]	0.04	[25]
Aitne	0.000003		1.5	[25]	2.6	22.7R	[25]	0.04	[25]
Eurydome	0.000003		1.5	[25]	2.6	22.7R	[25]	0.04	[25]
Euanthe	0.000003		1.5	[25]	2.6	22.8R	[25]	0.04	[25]
Euporie	0.000001		1.0	[25]	2.6	23.1R	[25]	0.04	[25]
Orthosie	0.000001		1.0	[25]	2.6	23.1R	[25]	0.04	[25]

Sponde	0.000001		1.0	[25]	2.6	23.0R	[25]	0.04	[25]
Kale	0.000001		1.0	[25]	2.6	23.0R	[25]	0.04	[25]
Pasithee	0.000001		1.0	[25]	2.6	23.2R	[25]	0.04	[25]
Hegemone	0.000003		1.5	[25]	2.6	22.8R	[25]	0.04	[25]
Mneme	0.000001		1.0	[25]	2.6	23.3R	[25]	0.04	[25]
Aoede	0.000006		2.0	[25]	2.6	22.5R	[25]	0.04	[25]
Thelxinoe	0.000001		1.0	[25]	2.6	23.5R	[25]	0.04	[25]
Arche	0.000003		1.5	[25]	2.6	22.8R	[25]	0.04	[25]
Kallichore	0.000001		1.0	[25]	2.6	23.7R	[25]	0.04	[25]
Helike	0.000006		2.0	[25]	2.6	22.6R	[25]	0.04	[25]
Carpo	0.000003		1.5	[25]	2.6	23.0R	[25]	0.04	[25]
Eukelade	0.000006		2.0	[25]	2.6	22.6R	[25]	0.04	[25]
Cyllene	0.000001		1.0	[25]	2.6	23.2R	[25]	0.04	[25]
Kore	0.000001		1.0	[25]	2.6	23.6R	[25]	0.04	[25]
Herse	0.000001		1.0	[25]	2.6	23.4R	[25]	0.04	[25]
S/2000 J11	0.000001		1.0	[25]	2.6	22.4R	[25]	0.04	[25]
S/2003 J2	0.000001		1.0	[25]	2.6	23.2R	[25]	0.04	[25]
S/2003 J3	0.000001		1.0	[25]	2.6	23.4R	[25]	0.04	[25]
S/2003 J4	0.000001		1.0	[25]	2.6	23.0R	[25]	0.04	[25]
S/2003 J5	0.000006		2.0	[25]	2.6	22.4R	[25]	0.04	[25]
S/2003 J9	0.0000001		0.5	[25]	2.6	23.7R	[25]	0.04	[25]
S/2003 J10	0.000001		1.0	[25]	2.6	23.6R	[25]	0.04	[25]
S/2003 J12	0.0000001		0.5	[25]	2.6	23.9R	[25]	0.04	[25]
S/2003 J15	0.000001		1.0	[25]	2.6	23.5R	[25]	0.04	[25]
S/2003 J16	0.000001		1.0	[25]	2.6	23.3R	[25]	0.04	[25]
S/2003 J18	0.000001		1.0	[25]	2.6	23.4R	[25]	0.04	[25]
S/2003 J19	0.000001		1.0	[25]	2.6	23.7R	[25]	0.04	[25]
S/2003 J23	0.000001		1.0	[25]	2.6	23.6R	[25]	0.04	[25]
S/2010 J1	0.000001		1.0	[60]	2.6	23.2r	[60]	0.04	[60]
S/2010 J2	0.000001		1.0	[60]	2.6	24.0r	[60]	0.04	[60]
S/2011 J1	0.000001		1.0	[61]	2.6	23.7R	[61]	0.04	[61]

S/2011 J2	0.000001		1.0	[61]	2.6	23.5R	[61]	0.04	[61]
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Saturnian System

Sat.	<u><i>GM</i></u> (km ³ /sec ²)		<u><i>Mean radius</i></u> (km)		<u><i>Mean density</i></u> (g/cm ³)	<u><i>Magnitude</i></u> <i>V</i> ₀ or <i>R</i>		<u><i>Geometric Albedo</i></u>	
Mimas	2.5026±0.0006	[27]	198.20±0.25	[28]	1.150±0.004	12.8	[29]	0.962±0.004	[59]
Enceladus	7.2027±0.0125	[27]	252.10±0.10	[28]	1.608±0.003	11.8	[29]	1.375±0.008	[59]
Tethys	41.2067±0.0038	[27]	533.00±0.70	[28]	0.973±0.004	10.2	[29]	1.229±0.005	[59]
Dione	73.1146±0.0015	[27]	561.70±0.45	[28]	1.476±0.004	10.4	[29]	0.998±0.004	[59]
Rhea	153.9426±0.0037	[27]	764.30±1.10	[28]	1.233±0.005	9.6	[29]	0.949±0.003	[59]
Titan	8978.1382±0.0020	[27]	2574.73±0.09	[55]	1.882±0.001	8.4	[29]	0.2	[29]
Hyperion	0.3727±0.0012	[27]	135.00±4.00	[54]	0.544±0.050	14.4	[29]	0.3	[29]
Iapetus	120.5038±0.0080	[27]	735.60±1.50	[28]	1.083±0.007	11	[29]	0.6	[29]
Phoebe	0.5532±0.0006	[27]	106.50±0.70	[54]	1.638±0.033	16.4	[34]	0.081±0.002	[30]
Janus	0.1263±0.0087	[54]	89.5±1.5	[54]	0.630±0.030	14.4	[29]	0.71±0.02	[59]
Epimetheus	0.0351±0.0047	[54]	58.1±1.8	[54]	0.640±0.062	15.6	[29]	0.73±0.03	[59]
Helene	0.00076		17.6±0.4	[54]	0.5	18.4	[29]	1.67±0.20	[59]
Telesto	0.00027		12.4±0.4	[54]	0.5	18.5	[29]	1.0	[29]
Calypso	0.00017		10.7±0.7	[54]	0.5	18.7	[29]	1.34±0.10	[59]
Atlas	0.00044±0.00015	[54]	15.1±1.2	[54]	0.460±0.110	19.0	[29]	0.4	[29]
Prometheus	0.01074±0.00285	[54]	43.1±2.7	[54]	0.480±0.090	15.8	[29]	0.6	[29]
Pandora	0.00924±0.00152	[54]	40.7±1.5	[54]	0.490±0.060	16.4	[29]	0.5	[29]
Pan	0.00033±0.00015	[54]	14.1±1.3	[54]	0.420±0.150	19.4	[32]	0.5	[32]
Methone	0.0000006		1.6±0.6	[54]	0.5	?		?	
Pallene	0.0000022		2.5±0.6	[54]	0.5	?		?	
Polydeuces	0.0000003		1.3±0.4	[54]	0.5	?		?	
Daphnis	0.0000052±0.0000052	[54]	3.8±0.8	[54]	0.340±0.260	?		?	
Anthe	0.0000001		0.9	[57]	0.5	?		?	
Aegaeon	0.000000004		0.3	[56]	0.5	?		?	
Ymir	0.00033		9	[33]	2.3	21.9R	[34]	0.06	[33]
Paaliaq	0.00055		11.0	[33]	2.3	21.1R	[34]	0.06	[33]

Tarvos	0.00018		7.5	[33]	2.3	22.7R	[34]	0.06	[33]
Ijiraq	0.000080		6	[33]	2.3	22.6R	[35]	0.06	[33]
Suttungr	0.000014		3.5	[33]	2.3	23.9R	[35]	0.06	[33]
Kiviuq	0.00022		8	[33]	2.3	22.1R	[34]	0.06	[33]
Mundilfari	0.000014		3.5	[33]	2.3	23.8R	[35]	0.06	[33]
Albiorix	0.0014		16	[33]	2.3	20.5R	[34]	0.06	[33]
Skathi	0.000021		4	[33]	2.3	23.6R	[35]	0.06	[33]
Erriapus	0.000051		5	[33]	2.3	23.4R	[34]	0.06	[33]
Siarnaq	0.0026		20	[33]	2.3	19.9R	[34]	0.06	[33]
Thrymr	0.000014		3.5	[33]	2.3	23.9R	[35]	0.06	[33]
Narvi	0.000023		3.5	[35]	2.3	23.8R	[35]	0.04	[35]
Aegir	0.000000		3.0	[35]	2.3	24.4R	[35]	0.04	[35]
Bebhionn	0.000000		3.0	[35]	2.3	24.1R	[35]	0.04	[35]
Bergelmir	0.000000		3.0	[35]	2.3	24.2R	[35]	0.04	[35]
Bestla	0.000000		3.5	[35]	2.3	23.8R	[35]	0.04	[35]
Farbauti	0.000000		2.5	[35]	2.3	24.7R	[35]	0.04	[35]
Fenrir	0.000000		2.0	[35]	2.3	25.0R	[35]	0.04	[35]
Fornjot	0.000000		3.0	[35]	2.3	24.6R	[35]	0.04	[35]
Hati	0.000000		3.0	[35]	2.3	24.4R	[35]	0.04	[35]
Hyrrokkin	0.000000		3.0	[35]	2.3	23.5R	[35]	0.04	[35]
Kari	0.000000		3.0	[35]	2.3	23.9R	[35]	0.04	[35]
Loge	0.000000		3.0	[35]	2.3	24.6R	[35]	0.04	[35]
Skoll	0.000000		3.0	[35]	2.3	24.5R	[35]	0.04	[35]
Surtur	0.000000		3.0	[35]	2.3	24.8R	[35]	0.04	[35]
Jarnsaxa	0.000000		3.0	[35]	2.3	24.7R	[35]	0.04	[35]
Greip	0.000000		3.0	[35]	2.3	24.4R	[35]	0.04	[35]
Tarqeq	0.000000		3.0	[35]	2.3	23.9R	[35]	0.04	[35]
S/2004 S7	0.000000		3.0	[35]	2.3	24.5R	[35]	0.04	[35]
S/2004 S12	0.000000		2.5	[35]	2.3	24.8R	[35]	0.04	[35]
S/2004 S13	0.000000		3.0	[35]	2.3	24.5R	[35]	0.04	[35]
S/2004 S17	0.000000		2.0	[35]	2.3	25.2R	[35]	0.04	[35]

S/2006 S1	0.000000		3.0	[35]	2.3	24.6R	[35]	0.04	[35]
S/2006 S3	0.000000		2.5	[35]	2.3	24.6R	[35]	0.04	[35]
S/2007 S2	0.000000		3.0	[35]	2.3	24.4R	[35]	0.04	[35]
S/2007 S3	0.000000		2.0	[35]	2.3	24.9R	[35]	0.04	[35]

Uranian Satellites

Sat.	<u><i>GM</i></u> (km ³ /sec ²)		<u><i>Mean radius</i></u> (km)		<u><i>Mean density</i></u> (g/cm ³)	<u><i>Magnitude</i></u> <i>V</i> ₀ or <i>R</i>		<u><i>Geometric Albedo</i></u>	
Ariel	86.4±5.0	[36]	578.9±0.6	[37]	1.592±0.092	13.70±0.04	[38]	0.39±0.04	[39]
Umbriel	81.5±5.0	[36]	584.7±2.8	[37]	1.459±0.092	14.47±0.04	[38]	0.21±0.02	[39]
Titania	228.2±5.0	[36]	788.9±1.8	[37]	1.662±0.038	13.49±0.04	[38]	0.27±0.03	[39]
Oberon	192.4±7.0	[36]	761.4±2.6	[37]	1.559±0.059	13.70±0.04	[38]	0.23±0.03	[39]
Miranda	4.4±0.4	[36]	235.8±0.7	[37]	1.214±0.109	15.79±0.04	[38]	0.32±0.03	[39]
Cordelia	0.0030		20.1±3.	[40]	1.3	23.62±0.35	[38]	0.07	[38]
Ophelia	0.0036		21.4±4.	[40]	1.3	23.26±0.25	[38]	0.07	[38]
Bianca	0.0062		27±2	[67]	1.3	22.52±0.24	[38]	0.065±0.010	[67]
Cressida	0.0229		41±2	[67]	1.3	21.58±0.11	[38]	0.069±0.007	[67]
Desdemona	0.0119		35±4	[67]	1.3	21.99±0.16	[38]	0.084±0.019	[67]
Juliet	0.0372		53±4	[67]	1.3	21.12±0.05	[38]	0.075±0.011	[67]
Portia	0.1122		70±4	[67]	1.3	20.42±0.05	[38]	0.069±0.008	[67]
Rosalind	0.0170		36±6	[67]	1.3	21.79±0.13	[38]	0.072±0.024	[67]
Belinda	0.0238		45±8	[67]	1.3	21.47±0.09	[38]	0.067±0.024	[67]
Puck	0.1931		81±2	[67]	1.3	19.75±0.05	[38]	0.104±0.006	[67]
Caliban	0.020		36	[42]	1.5	22.4R	[42]	0.04	[42]
Sycorax	0.18		75	[42]	1.5	20.8R	[42]	0.04	[42]
Prospero	0.0066		25	[42]	1.5	23.2R	[42]	0.04	[42]
Setebos	0.0058		24	[42]	1.5	23.3R	[42]	0.04	[42]
Stephano	0.0017		16	[42]	1.5	24.1R	[42]	0.04	[42]
Trinculo	0.00031		9	[42]	1.5	25.4R	[42]	0.04	[42]
Francisco	0.00056		11	[42]	1.5	25.0R	[42]	0.04	[42]
Margaret	0.00042		10	[42]	1.5	25.2R	[42]	0.04	[42]
Ferdinand	0.00042		10	[42]	1.5	25.1R	[42]	0.04	[42]

Perdita	0.0012		13±1	[67]	1.3	23.6V	[40]	0.070±0.006	[67]
Mab	0.0006		12±1	[67]	1.3	24.6V	[67]	0.103±0.006	[67]
Cupid	0.0002		9±1	[67]	1.3	25.8V	[67]	0.070±0.006	[67]

Neptunian Satellites

Sat.	<u><i>GM</i></u> (km ³ /sec ²)		<u><i>Mean radius</i></u> (km)		<u><i>Mean density</i></u> (g/cm ³)	<u><i>Magnitude</i></u> <i>V</i> ₀ or <i>R</i>		<u><i>Geometric Albedo</i></u>	
Triton	1427.6±1.9	[43]	1353.4±0.9	[44]	2.059±0.005	13.54	[46]	0.719	[46]
Nereid	2.06		170.±25.	[47]	1.5	19.2R	[48]	0.155	[47]
Naiad	0.013		33.±3.	[48]	1.3	23.91	[48]	0.072	[48]
Thalassa	0.025		41.±3.	[48]	1.3	23.32	[48]	0.091	[48]
Despina	0.14		75.±3.	[48]	1.3	22.00	[48]	0.090	[48]
Galatea	0.25		88.±4.	[48]	1.3	21.85	[48]	0.079	[48]
Larissa	0.33		97.±3.	[48]	1.3	21.49	[48]	0.091	[48]
Proteus	3.36		210.±7.	[48]	1.3	19.75	[48]	0.096	[48]
Halimede	0.012		31.0	[49]	1.5	24.5R	[49]	0.04	[49]
Psamathe	0.0033		20.0	[49]	1.5	25.5R	[49]	0.04	[49]
Sao	0.0045		22.0	[49]	1.5	25.5R	[49]	0.04	[49]
Laomedeia	0.0039		21.0	[49]	1.5	25.5R	[49]	0.04	[49]
Neso	0.011		30.0	[49]	1.5	24.6R	[49]	0.04	[49]
S/2004 N1	0.0003		9.0	[62]	1.3	26.5V	[62]	0.10	[62]

Pluto's Satellites

Sat.	<u><i>GM</i></u> (km ³ /sec ²)		<u><i>Mean radius</i></u> (km)		<u><i>Mean density</i></u> (g/cm ³)	<u><i>Magnitude</i></u> <i>V</i> ₀ or <i>R</i>		<u><i>Geometric Albedo</i></u>	
Charon	102.3±0.2	[66]	603.6±1.4	[51]	1.664±0.012	17.26±0.01	[52]	0.372±0.012	[52]
Nix	0.0013±0.0023	[66]	23.0±2.	[63]	2.1	23.4V±0.2	[63]	0.35	[63]
Hydra	0.0065±0.0022	[66]	30.5±4.	[63]	0.8	22.9V±0.1	[63]	0.35	[63]
Kerberos	0.0011±0.0005	[66]	14.0	[64]	1.4	26.1V±0.3	[64]	0.35	[64]
Styx	0.0000±0.0008	[66]	10.0	[65]	?	27.0V±0.3	[65]	0.35	[65]

Table Column Headings

GM	GM: $G = 6.67259 \times 10^{-20} (\text{kg}^{-1} \text{ km}^3 \text{ sec}^{-2})$, $M = \text{mass (kg)}$, if the density is assumed (i. e. has no uncertainty) the value is computed from the density and mean radius
Mean radius	Mean radius: if no uncertainty is given, the value is computed from the magnitude and albedo
Mean density	Mean density: derived from the GM and mean radius when an uncertainty is given, otherwise assumed
Magnitude	Mean opposition magnitude: V_0 , or red magnitude, R
<u>Geometric Albedo</u>	<u>Geometric albedo</u>

References

1. Standish, E. M. 1998. JPL Planetary and Lunar Ephemerides, DE405/LE405.
2. Bills, B.G. and Ferrari, A.J. 1977. "A Harmonic Analysis of Lunar Topography", *Icarus* **31**, 244-259.
3. Explanatory Supplement to the Astronomical Almanac 1992. (P. K. Seidelmann Ed.) University Science Books, Mill Valley, CA.
4. Jacobson, R. A. 2010. "The Orbits and Masses of the Martian Satellites and the Libration of Phobos", *AJ* **139**, 668-679.
5. Thomas, P.C. 1989b. "Shapes of Small Satellites", *Icarus* **77**, 248-274.
6. Zellner, B., and Capen, R. C. 1974. "Photometric Properties of the Martian Satellites", *Icarus* **23**, 437-444.
7. Thomas, P. C., Adinolfi, D., Helfenstein, P., Simonelli, D. and Veverka, J. 1996. "The Surface of Deimos: Contribution of Materials and Processes to Its Unique Appearance", *Icarus* **123**, 536-556.
8. Jacobson, R. A. 2003b. private communication of constants used in the JUP230 ephemeris
9. Thomas, P. C., Davies, M. E., Colvin, T. R., Oberst, J., Schuster, P., Neukum, G., Carr, M. H., McEwen, A., Schubert, G., Belton, M. J. S., and the Galileo Imaging Team 1998, "The Shape of Io from Galileo Limb Measurements", *Icarus* **135**, 175-180.
10. Thomas, P.C. 2002. private communication of limb fit results
11. Anderson, J. D., Jacobson, R. A., Lau, E. L., Moore, W. B., Olsen, O., Schubert, G., and Thomas, P. C. 2000. "Shape, Mean Radius, Gravity Field and Interior Structure of Ganymede", *Bulletin of the American Astronomical Society* **33**(3), 1101.
12. Anderson, J.D., Jacobson, R.A., McElrath, T. P., Schubert, G., Moore, W.B., and Thomas, P. C. 2000a. "Shape, Mean Radius, Gravity Field and Interior Structure of Callisto", *Icarus* **153**, 157-161.
13. Thomas, P.C. 2002. private communication of Amalthea volume based on its shape model
14. Morrison, D., N. D. Morrison, and Lazarewicz, A. R. 1974. "Four color photometry of the Galilean satellites", *Icarus* **23**, 399-416.
15. Thomas, P. and Veverka, J. 1982. "Amalthea", In Satellites of Jupiter (D. Morrison Eds.) pp. 147-173. Univ. of Arizona Press, Tucson.
16. Simonelli, D. P. and Veverka, J. 1984. "Voyager Disk-Integrated Photometry of Io", *Icarus* **59**, 406-425.
17. Buratti, B. and Veverka, J. 1983. "Voyager Photometry of Europa", *Icarus* **55**, 93-110. (average of leading and trailing hemisphere values)
18. Morrison, D. and Morrison, N. D. 1977. "Photometry of the Galilean Satellites", In Planetary Satellites (J. A. Burns, Ed.) pp. 363-378. Univ. of Arizona Press, Tucson.
19. Simonelli, D. P., Rossier, L., Thomas, P. C., Veverka, J., Burns, J. A., and Belton, M. J. S. 2000. "Leading/Trailing Albedo Asymmetries of Thebe, Amalthea, and Metis", *Icarus* **147**, 353-365.
20. Rettig, T. W. and Walsh, K. W. 2001. "Implied Evolutionary Differences of the Jovian Irregular Satellites

- from a BVR Color Survey", presented at Jupiter: Planet, Satellites, and Magnetosphere, Boulder Colorado
21. Thomas, P. C., Burns, J. A., Rossier, L., Simonelli, D., Veverka, J., Chapman, C. R., Klaasen, K., Johnson, T. V., and Belton, M. J. S. 1998. "The Small Inner Satellites of Jupiter", *Icarus* **135**, 360-371.
 22. Veverka, J., Thomas, P., and Synnott, S. 1982. "The Inner Satellites of Jupiter", *Vistas in Astronomy* **25**, 245-262.
 23. Gladman, B. J. 2000. *IAU Circular* 7472
 24. Jewitt, D. C. 2001. personal communication
 25. Sheppard, S. S. 2010. <http://www.dtm.ciw.edu/users/sheppard/satellites/jupsatdata.html>
 26. Rettig, T. W., Walsh, K., and Consolmagno, G. 2001. "Implied Evolutionary Differences of the Jovian Irregular Satellites from a BVR Color Survey", *Icarus* **154**, 313-320.
 27. Jacobson, R. A., Antreasian, P. G., Ardalan, S., Criddle, K. E., Ionasescu, R., Jones, J. B., Parcher, D., Pelletier, F. J., Roth, D. C., Thompson, P., Vaughan, A. 2008. "The Gravity Field of the Saturnian System and the Orbits of the Major Saturnian satellites", Presented at the Saturn After Cassini-Huygens Symposium held at Imperial College London, UK
 28. Thomas, P. C., Burns, J. A., Helfenstein, P., Squyres, S., Veverka, J., Porco, C., Turtle, E. P., McEwen, A., Denk, T., Giese, B., Roatsch, T., Johnson, T. V., Jacobson, R. A. 2007. "Shapes of the Saturnian icy satellites and their significance", *Icarus* **190**, 573-584.
 29. Morrison, D., Johnson, T. V., Shoemaker, E. M., Soderbloom, L. A., Thomas, P., Veverka, J., and Smith, B. A. 1984. "Satellites of Saturn: Geological Perspective", In Saturn (T. Gehrels and M. S. Matthews Eds.) pp. 609-639. Univ. of Arizona Press, Tucson.
 30. Simonelli, D. P., Kay, J., Adinolfi, D., Veverka, J., Thomas, P. C., and Helfenstein, P. 1999. "Phoebe: Albedo Map and Photometric Properties", *Icarus* **138**, 249-258.
 31. Jacobson, R. A., Spitale, J., Porco, C. C., Beurle, K., Cooper, N. J., Evans, M. W., Murray, C. D. 2008. "Revised Orbits of Saturn's Small Inner Satellites", *AJ* **135**, 261-263.
 32. Porco, C. C., Weiss, J. W., Thomas, P. C., Richardson, D. C., Jacobson, R. A., Spitale, J. 2006. "Physical Characteristics and Possible Accretionary Origins for Saturn's Small Satellites", Presented at the 37th Lunar and Planetary Science Conference, League City, TX
 33. Nicholson, P. D. 2001. private communication
 34. Grav, T., Holman, M. J., Gladman, B. J., and Aksnes, K. 2003. "Photometric survey of the irregular satellites", *Icarus* **166**, 33-45.
 35. Sheppard, S. S. 2010. <http://www.dtm.ciw.edu/users/sheppard/satellites/satsatdata.html>
 36. Jacobson, R. A. 2007. "The Gravity Field of the Uranian System and the Orbits of the Uranian Satellites and Rings", *BAAS* **39**, 453.
 37. Thomas, P.C. 1988. "Radii, shapes, and topography of the satellites of Uranus from limb coordinates", *Icarus* **73**, 427-441.
 38. Karkoschka, E. 2001a. "Comprehensive Photometry of the Rings and 16 Satellites of Uranus with the Hubble Space Telescope", *Icarus* **151**, 51-68.
 39. Veverka, J., Brown, R. H., and Bell, J. F. 1991. "Uranus Satellites: Surface Properties", In Uranus (J. T. Bergstrahl, E. D. Miner, and M. S. Matthews Eds.) pp. 528-560. Univ. of Arizona Press, Tucson.
 40. Karkoschka, E. 2001b. "Voyager's Eleventh Discovery of a Satellite of Uranus and Photometry and the First Size Measurements of Nine Satellites", *Icarus* **151**, 69-77
 41. Thomas, P.C., Weitz, C., and Veverka, J. 1989a. "Small satellites of Uranus Disk-integrated photometry and estimated radii", *Icarus* **81**, 92-101.
 42. Sheppard, S. S., Jewitt, D., Kleyna, J. 2005. "An Ultradeep Survey for Irregular Satellites of Uranus: Limits to Completeness", *Astronomical Journal* **129**, 518-525.
 43. Jacobson, R. A. 2009. "The Orbits of the Neptunian Satellites and the Orientation of the Pole of Neptune",

- AJ* **137**, 4322.
44. Thomas, P. C. 2000. "The Shape of Triton from Limb Profiles", *Icarus* **148**, 587-588.
 45. Jacobson, R. A. 2008. "Reconstruction of the Voyager~2 Neptune Encounter in the ICRF System", AIAA Paper~2008-7372 presented at the AIAA/AAS Astrodynamics Specialist Conference, Honolulu, HI
 46. Hicks, M. D, and Buratti, B. J. 2004. "The spectral variability of Triton from 1997--2000", *Icarus* **171**, 210-218.
 47. Thomas, P., Veverka, J., and Helfenstein, P. 1991b. "Voyager Observations of Nereid", *Journal of Geophysical Research* **96**(19), 253-19, 259.
 48. Karkoschka, E. 2003. "Sizes, shapes, and albedos of the inner satellites of Neptune", *Icarus* **162**, 400-407
 49. Sheppard, S. S., Jewitt, D., and Kleyna, J. 2006. "A Survey for Normal Irregular Satellites around Neptune: Limits to Completeness", *Astronomical Journal* **132**, 171-176.
 50. Jacobson, R. A. 2007. private communication of constants used in the PLU017 ephemeris
 51. Sicardy, B., et al. 2006. "Charon's size and an upper limit on its atmosphere from a stellar occultation", *Nature* **439**, 52-54.
 52. Reinsch, K., Burwitz, V., and Festou, M. C. 1994. "Albedo Maps of Pluto and Improved Physical Parameters of the Pluto-Charon System", *Icarus* **108**, 209-218.
 53. Tholen, D.~J., Buie, M.~W., Grundy, W.~M., Elliott, G.~T. 2008. "Masses of Nix and Hydra", *AJ* **135**, 777-784.
 54. Thomas, P. C. 2010. "Sizes, shapes, and derived properties of the saturnian satellites after the Cassini nominal mission", *Icarus* **208**, 395-401.
 55. Zebker, H. A. and Stiles, B. and Hensley, S. Lorenz, R. and Kirk, R. L. and Lunine, J. 2009. "Size and Shape of Saturn's Moon Titan", *Science* **324**, 921-923.
 56. Hedman, M. M., Cooper, N. J., Murray, C. D., Beurle, K., Evans, M. W., Tiscareno, M. S., and Burns, J. A. 2010. "Aegaeon (Saturn LIII), a G-ring object", *Icarus* **207**, 433-447.
 57. Cooper, N. J., Murray, C. D., Evans, M. W., Beurle, K., Jacobson, R. A., and Porco, C. C. 2008. "Astrometry and dynamics of Anthe (S/2007 S4), a new satellite of Saturn", *Icarus* **195**, 765-777.
 58. Weaver, H. A., et al. 2005. *IAU Circular* 8625
 59. Verbiscer, A., French, R., Showalter, M., and Helfenstein, P. 2007. "Enceladus: Cosmic Graffiti Artist Caught in the Act", *Science* **315**, 815.
 60. Alexandersen, M., Gladman, B., Veillet, C., Jacobson, R., Brozovic, M., and Rousselot, P. 2012. "Discovery of two additional Jovian irregulars", *Astronomical Journal* **144**, 21.
 61. Sheppard, S. S. 2012. "New Satellites of Jupiter: S/2011 J 1 and S/2011 J 2", *Central Bureau Electronic Telegrams* **3002**
 62. Showalter, M. R., de Pater, I., Lissauer, J. J., French, R. S. 2013. "New Satellite of Neptune: S/2004 N 1", *Central Bureau Electronic Telegrams* **3586**
 63. Weaver, H. A., Stern, S. A., Mutchler, M. J., Steffl, A. J., Buie, M. W., Merline, W. J., Spencer, J. R., Young, E. F., and Young, L. A. 2006. "Discovery of two new satellites of Pluto", *Nature* **439**, 943.
 64. Showalter, M. R., Hamilton, D. P., Stern, S. A., Weaver, H. A., Steffl, A. J., and Young, L. A. 2011. "New Satellite of (134340) Pluto: S/2011 (134340) 1", *IAU Circular* **9221**
 65. Showalter, M. R., Weaver, H. A., Stern, S. A., Steffl, A. J., Buie, M. W., Merline, W. J., Mutchler, M. J., Soummer, R., and Throop, H. B. 2012. "New Satellite of (134340) Pluto: S/2012 (134340) 1", *IAU Circular* **9253**
 66. Brozovic, M. and Jacobson, R. A. 2013. "The Orbits, Masses of Pluto's Satellites", Presented at The Pluto System on the Eve of Exploration by New Horizons: Perspectives, Predictions held at APL, Laurel, MD
 67. Showalter, M. R. and Lissauer, J. J. 2006. "The Second Ring-Moon System of Uranus: Discovery and Dynamics", *Science* **311**, 973.

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